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**College of Business, Arts and Social Sciences**

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| <b>Student Number:</b> | 2501536                                                                                                                                              |
| <b>Module Code:</b>    | MG5636                                                                                                                                               |
| <b>Module Title:</b>   | Dissertation                                                                                                                                         |
| <b>Project Title:</b>  | Automated Financial Analysis and Artificial Intelligence: An Analysis of Strategic Decision Making within Management Accounting using Secondary Data |
| <b>Supervisor:</b>     | Dr. Grigorios Theodosopoulos                                                                                                                         |

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**Automated Financial Analysis and Artificial  
Intelligence:  
An Analysis of Strategic Decision Making within  
Management Accounting Using Secondary Data**

**Student ID: 2501536**

## Abstract

The increasing integration of Artificial Intelligence (AI) within finance functions has transformed the analytical capabilities of Strategic Management Accounting (SMA). Financial Automation Analysis (FAA) enables organisations to automate forecasting, scenario modelling and performance monitoring, creating new opportunities for data-driven strategic decision-making. Despite rapid adoption, existing research has largely focused on technical capabilities rather than the broader organisational and strategic implications of these technologies.

This study investigates how and why the FAA influences strategic decision-making in management accounting. A **theory-driven secondary data analysis** is employed, examining peer-reviewed academic literature, professional reports, and organisational case evidence primarily published between 2019 and 2025. The analysis is guided by three theoretical perspectives: **Decision Theory, Socio-Technical Systems Theory and the Dynamic Capabilities Framework**. Four analytical propositions are developed to examine relationships between FAA adoption, decision quality, governance and trust, professional transformation and organisational agility.

The findings demonstrate that FAA improves forecasting accuracy, analytical insight and decision speed when implemented as a decision support system supported by human judgement and effective governance structures.

However, its strategic value depends on governance maturity, professional capability development and organisational alignment. The study concludes that FAA represents a **socio-technical transformation** whose strategic impact is contingent upon organisational context and the maturity of its governance and professional capabilities.

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## **Chapter 1 – Introduction**

### **1.1 Overview of the Chapter**

This chapter introduces the context, motivation and structure of the dissertation, "Automated Financial Analysis and Artificial Intelligence: An analysis of strategic decision making within management accounting using secondary data." The chapter introduces the growing adoption of Artificial Intelligence (AI) and Financial Automation Analysis (FAA) in Strategic Management Accounting (SMA), and outlines the research problem, aim, objectives and questions.

### **1.2 Background and Context**

Over the last ten years, digital technology has significantly changed organisational finance functions and how financial information is generated and used for strategic decision-making and analysis. AI-enabled technologies are being integrated into management accounting systems to assist with complex analytical tasks and provide forward-looking insights (Cao and Zhang, 2025; Fehrenbacher, Ghio and Weisner, 2023).

One major piece of evidence for the change in how organisations use their management accounting systems is the development of Financial Automation Analysis (FAA). FAA is described by Mahlendorf et al. (2023) as the use of AI-driven analytical tools to automatically generate budgets, forecasts, variance analysis, performance dashboards and scenario modelling.

Unlike traditional rule-based automation systems, FAA systems can learn from both structured and unstructured data, enabling continuous forecasting and near-real-time reporting (Mahlendorf, Martin and Smith, 2023). As a result of implementing FAA, organisations have reported reduced reporting cycle times and increased speed, breadth, and accessibility of financial information available to strategic decision-makers (Rieg, 2025).

Industry reports indicate that AI-enabled financial analytics are increasingly integrated into finance transformation initiatives across multiple sectors (ACCA, 2024; Deloitte, 2025). For example, ACCA (2024) reports a shift away from periodic reporting toward continuous performance monitoring, enabling

organisations to respond more quickly to emerging strategic risks and opportunities.

Deloitte (2025) reports that increasing AI adoption in finance functions is expected to shift the role of finance from operational processing toward strategic value creation. Although research has demonstrated that many companies are overestimating the strategic value of artificial intelligence, as simply having the technical capabilities to use artificial intelligence does not necessarily mean that an organisation is prepared to effectively utilise this technology for making decisions (Raisch and Krakowski, 2021).

Case study evidence from individual organisations further supports this trend. The partnership between J Sainsbury PLC and Microsoft provides an example of how AI-enabled forecasting tools can be used to improve demand planning and strategic responsiveness in a large retail organisation (J Sainsbury PLC and Microsoft Corporation, 2024).

Although these examples demonstrate the potential of the FAA to support improved strategic decision-making outcomes, they also illustrate the challenges related to governance, transparency and developing the professional capabilities of employees.

### **1.3 Research Problem and Gap**

Although the adoption of FAA technology is spreading rapidly, there is currently a lack of integrated knowledge about its implications at a strategic level for management accounting.

While there have been many studies about both the technological aspects of using Artificial Intelligence (AI) enabled analytics, and what can be expected from digital transformation in accounting, there are three areas that have yet to be adequately addressed by researchers.

Firstly, while many of the studies published about AI and digital transformation have examined the ability of algorithms to perform tasks, very few have looked at the impact of such tools on organisational strategy development processes.

Secondly, most of the current studies have examined governance, professional transformation and analytical capability separately rather than examining them as an interactive set of organisational systems.

Lastly, there is little empirical evidence available regarding how these factors interact with each other to affect strategic decision making; therefore, the overall effect of all three on strategic decision making is still largely unexplored.

Further, most of the research previously conducted was of a conceptual nature or was based upon specific empirical contexts, which has limited the generalizability of findings related to the extent to which FAA improves or hinders the effectiveness of strategic decisions (Arnold and Sutton, 2019).

Therefore, it is unclear as to why certain organisations can improve their forecasting accuracy, speed up their decision cycle, and increase their ability to respond strategically after implementing FAA, while other organisations experience difficulties with the implementation process, ethical dilemmas, and opposition from professional employees (Lehner et al., 2022).

The limitations created by primary data collection also compound this limitation. It can be difficult to obtain access to senior finance personnel and management accountants because of confidentiality concerns and the demands of their workload, making large-scale interviews or surveys impractical.

These limitations highlight the need for a theory-based secondary data analysis using academic literature, professional reports, and organisational case studies to provide additional insight to help explain the strategic implications of adopting FAA.

As such, this study will address the following overall research question:

**How and why does financial automation analysis affect the way strategic decisions are made in management accounting in various types of organisations?**

## **1.4 Research Aim, Objectives, and Questions**

### **Research Aim**

The overall aim of this thesis is to analyse the impact of Financial Automation Analysis (FAA) and Artificial Intelligence (AI) on the strategic decision-making process in management accounting through an examination of the literature from academia, practice and organisations.

## **Research Objectives**

To meet the above aim, the research will address four main objectives:

1. To examine how AI-enabled automation influences decision quality, analytical speed, and strategic relevance within management accounting.
2. To assess the ethical and governance structures that develop trust and adoption of FAA systems.
3. To study how FAA adoption changes the professional roles and skills of management accountants.
4. To assess how the FAA contributes to organisational agility and strategic performance within dynamic environments.

## **Research Questions**

Through these four objectives, the research will seek to answer the following research questions:

1. How do the results of Financial Automation Analysis (FAA) improve the quality and effectiveness of strategic decision-making within Strategic Management Accounting?
2. What ethical and governance issues arise when there is increased reliance on AI-driven financial analysis?
3. In what ways are the roles and capabilities of management accountants developing because of FAA adoption?
4. In what way can the FAA contribute to an organisation's ability to be responsive and agile, and its strategic performance?
5. Which of the various analytical frameworks is most effective at explaining the inter-relationship between FAA, governance, decision quality and organisational capabilities?

Theoretical framework's usefulness (effectiveness) can be measured through three different dimensions: how well a theoretical framework explains, how well it accounts for contingencies that may exist within an organisation, and whether they are supported by the empirical data from this research.

## **1.5 Significance of the Study**

### **Academic Significance**

This study contributes to the literature by integrating decision theory, socio-technical systems theory and dynamic capabilities to explain the strategic implications of FAA.

Literature has identified various types of risks associated with technology becoming too dominant, including dependence on automation (Arnold and Sutton, 2019) and ethically concerning issues with decision-making due to the use of AI (Martin, Shilton and Smith, 2020).

In addition to identifying new knowledge related to the risks of FAA as a socio-technical phenomenon, the study also identifies how FAA could be used strategically as opposed to just being viewed as a technological innovation.

### **Practical Significance**

The practical implications of this study are for Chief Financial Officers (CFO's), Finance Managers, and Senior Decision-Makers who may utilise FAA to create strategic advantages through their organisations' use of FAAs.

The evidence provided by this research indicates that the strategic advantages of AI-enabled Finance Functions will include enhanced forecasted accuracy and enhanced decision speed; however, these advantages are dependent upon appropriate governance arrangements and the development of professional capabilities (EY, 2025; ACCA, 2025).

### **Professional Significance**

The Professional Implications of the FAA are that they represent a significant shift in the required competencies of Management Accountants. Management Accountants are now expected to function as strategic advisors to the organisation and provide context to the organisation for AI-generated information, rather than simply providing financial reports (Quattrone, 2020).

The secondary data utilised within this study provides additional support to the debate of what skills will be required by Management Accountants in the future and what professional development priorities should be established for

Management Accountants based on future required skills, which is a priority for many professional bodies, including ACCA (2024).

### **Methodological Significance**

The Methodological Implications of this study are that it supports the viability of utilising theory-driven secondary data analysis as an empirically valid methodological approach for conducting Accounting Research.

Through the utilisation of secondary data from three different sources: academic studies, professional reports, and organisational cases, the study provides a methodologically rigorous examination of FAA without having to collect primary data; therefore, it addresses many of the common feasibility concerns when conducting postgraduate research.

## **1.6 Methodological Overview**

This research utilises a theory-driven secondary data analysis approach based on three main types of data sources:

1. Peer-reviewed academic journals
2. Professional/Institutional Reports
3. Organisational Case Studies

Academic literature provides the theoretical bases and analytical frameworks for this study (Warner and Wäger, 2019; Csaszar et al., 2024) as well as professional reports providing evidence at an industry level of the adoption of the FAA by firms, their governance processes, and performance results (Thomson Reuters Institute, 2024; OECD, BCG and INSEAD, 2025). Examples of organisational cases are Sainsbury's and Grant Thornton, which provide contextualised evidence of the implementation of the FAA and the strategic outcomes of that implementation (J Sainsbury plc and Microsoft Corporation, 2024; Dayshape, 2025). To identify the relationships between decision quality, governance, professional transformation and organisational agility, the study will use both qualitative thematic analysis and descriptive quantitative comparative methods.

## **1.7 Structure of the Dissertation**

The dissertation will be divided into five chapters, outlined below:



**Chapter One:** Introduction: This chapter will provide an overview of the research background; define the research problem; outline the research aims, objectives and methodology; and discuss the importance and relevance of the study.

**Chapter Two:** Literature Review: In this chapter, a critical review of all relevant published research relating to the four main themes of the dissertation (FAA, AI, Governance and Strategic Decision Making) will be conducted. The purpose of this review is to develop several analytical propositions that can be used to guide and support the subsequent stages of the research.

**Chapter Three:** Methodology: This chapter will explain and justify the research philosophy adopted by the study; describe the research design and data collection methods; identify and describe the data sources and analytical techniques that were employed to collect and analyse the data.

**Chapter Four:** Findings and Analysis: This chapter will present and analyse the evidence collected during the study from both secondary data sources and organisational case studies.

**Chapter Five:** Conclusions and Recommendations – The final chapter draws together the main research results and reflects on what they signify; provides conclusions as to how the study has a direct impact on all levels of organisations that are responsible for strategic decisions; outlines the limitations of the research; and identifies potential avenues for additional research or best practice for strategic decision-making.

## **Chapter 2 – Literature Review**

### **2.1 Introduction**

The increasing use of Artificial Intelligence (AI) and Financial Automation Analysis (FAA) for Strategic Management Accounting (SMA) purposes has received considerable attention from academics and practitioners alike (Baldwin, Brown and Trinkle, 2006; Ucoglu, 2020).

Early studies in Digital Accounting Systems have demonstrated a significant increase in the use of big data and analytics in the accounting practice

(Vasarhelyi, Kogan and Tuttle, 2015; Bhimani and Willcocks, 2014; Davenport and Ronanki, 2018; Tiron-Tudor and Deliu, 2021).

Historically, the design of management accounting systems was to support periodic reporting, cost control and retrospective evaluation of performance. However, recent literature critiques traditional management accounting systems for their inability to support strategic decision making in volatile and unpredictable business environments (Cao and Zhang, 2025; Quattrone, 2020).

Therefore, AI-enabled FAA technologies (e.g. Machine Learning, Predictive Analytics, Deep Learning and Advanced Data Visualisation) have been implemented to perform routine analytical tasks faster than humans can do them, to enhance forecasting, scenario modelling, and to monitor performance in real time (Rieg, 2025; Mahlendorf, Martin and Smith, 2023).

Although the FAA may improve decision quality and organisational agility, empirical evidence remains mixed due to governance challenges, professional resistance and implementation differences. (Arnold and Sutton, 2019; Lehner et al., 2022).

The inconsistency in the empirical results indicates that the strategic effects of FAA cannot be understood based on technological capability alone; it must be studied using a comprehensive theoretical model.

Therefore, this chapter provides a critical review of relevant literature through three theoretically complementary lenses: Decision Theory, Socio-Technical Systems (STS) Theory, and the Dynamic Capabilities Framework. Each of the three theoretical frameworks will be applied to five conceptually related domains of investigation: decision quality, technological integration, ethical governance and trust, professional transformation and organisational agility. The chapter concludes by providing a conceptual framework for empirical study and a set of analytical propositions (P1-P4) that guide the empirical research.

## **2.2 Decision Theory, Bounded Rationality, and AI-Automated Decision-Making (P1)**

Decision-making lies at the core of management accounting, particularly in strategic contexts characterised by uncertainty, trade-offs, and long-term consequences. Decision-making in classical decision theory assumes that

decision makers are rational and able to analyse and compare all available information to make an optimal decision.

However, the theory of bounded rationality challenges this assumption by recognising cognitive limitations that constrain individuals' ability to process large volumes of complex data.

Within management accounting, bounded rationality often results in reliance on heuristics, simplified models, and historical data, limiting strategic responsiveness. AI-based FAA systems are therefore increasingly framed as mechanisms capable of mitigating bounded rationality by performing data-intensive analytical tasks and generating predictive insights at scale (Arnold and Sutton, 2019). Decision support systems powered by AI improve the decision-making process for managers by examining extensive data sets within organisations (Brynjolfsson and McAfee, 2017).

The emergence of research also indicates that the reduction of bounded rationality through automation does not occur uniformly, nor is it without risk. Training algorithms for historical data may include structural bias inherent within that data; therefore, algorithms can either reproduce or amplify existing distortions as opposed to reducing them (Martin et al., 2020). An increase in analytical ability does not necessarily equate to an improvement in strategic judgment; decision quality will continue to depend on interpretability, the presence of governance safeguards, and the continued engagement by humans who maintain a level of critical thinking with respect to their use of automated tools. Therefore, while the FAA has the potential to positively affect decision quality, its effect is dependent on social/technical alignment rather than simply the technology itself.

However, this optimistic perspective is problematised by research on technology dominance. Arnold and Sutton (2019) argue that intelligent systems may both enhance and undermine decision quality. On one hand, automation can lead to consistent and accurate analysis; on the other hand, automation may cause managers to rely too heavily on system output and reduce the managers' level of engagement and situational awareness, especially in management accounting, since strategic decisions involve context specific interpretations, ethical

judgments, and organizational knowledge, much of which cannot be formally documented or captured.

Studies have empirically illustrated the tension between analytical performance and managerial interpretation. For example, Fehrenbacher, Ghio and Weisner (2023) found that predictive analytics significantly influenced managerial recommendations based on their perception of credibility and interpretability of outputs. While the FAA has reduced forecast variability and improved analytical consistency, the impact of the FAA on decision quality will depend on how trusting users are and whether they can critically assess the automated outputs.

Similarly, Giantsidi and Tarantola (2025) demonstrate that deep learning models outperform traditional statistical methods in financial forecasting, yet caution that model opacity can undermine interpretability and managerial confidence.

However, while many studies have examined the technical forecasting accuracy of systems like those used by the FAA, very few have investigated the impact of such systems on manager judgment in an organisational context.

Collectively, this literature suggests that FAA enhances decision quality only when used as a decision-support mechanism, rather than a substitute for managerial judgment.

This conditional relationship underpins Proposition P1, which posits that improvements in decision quality arise when automation is complemented by active human oversight.

### **2.3 Socio-Technical Systems Theory and the Transformation of Management Accounting (P1 & P3)**

Socio-Technical Systems (STS) theory provides a critical framework for understanding the organisational implications of FAA adoption. STS theory emphasises that technological systems cannot be evaluated independently of the social systems within which they operate. Organisational outcomes emerge from interactions between technical tools, professional roles, skills, norms, and power relations.

FAA can therefore be interpreted as a socio-technical transformation rather than merely a technological upgrade. The concept of AI is seen by Jarrahi (2019) as enabling a form of human-AI symbiosis where AI-based automation enables the

decision-making process, and humans continue to make the interpretations, ethical judgments and strategic decisions. This view is very similar to the changing role of Management Accountants from being transactional reporters to strategic advisors.

Quattrone (2020) extends this argument by demonstrating that digitalisation increases, rather than reduces, the complexity of management accounting work. Digital systems produce new kinds of information which require interpretation, negotiation, and legitimation in organisational settings. Consequently, as digital systems become more prevalent and pervasive, management accountants are increasingly taking on hybrid roles that include analytic, strategic and communication skills.

Empirical evidence exists to support this transformation. Mahlendorf, Martin and Smith (2023) found that the automation of reporting tasks redirected the accountants' time from reporting to analysis and collaborative working across departments. However, Mahlendorf et al. also emphasised that a strategic benefit of automation occurs only when professionals are proactively examining the results produced by the automation. Rieg (2025) also demonstrated that digital dashboards improve performance only when users actively interact with the analytical data contained in the dashboard; passive consumption of the dashboard results in limited improvements.

Without complementary changes in skills, roles, and organisational culture, the benefits of automation are unlikely to be fully realised.

## **2.4 Technology Integration and Data Infrastructure**

Technological integration is a major determinant of the effectiveness of the Financial Automation Analysis (FAA) since only when analytical tools are embedded in an organisation's decision-making processes will they be able to generate strategic value (Appelbaum, Kogan and Vasarhelyi, 2017). Although early research on AI and accounting has been heavily reliant on algorithmic sophistication, the current body of knowledge has moved toward recognising that only when analytics are embedded within an organisation's processes and decision-making routines will they be able to generate strategic value (Barreto et al., 2025). Management accountants rely upon reliable data from ERP systems,

operational databases, and external markets (Appelbaum, Kogan and Vasarhelyi, 2017).

Fragmented data architecture limits the reliability of analysis and, therefore, the strategic use of FAA output (Mahlendorf, Martin and Smith, 2023). Barreto et al. (2025) identify that the value of analytics can arise from processes which provide mechanisms to coordinate, learn, and control. As such, organisations that treat analytics as separate tools do not typically realise long-term performance benefits. On the other hand, organisations that can incorporate FAA into budgeting, forecasting, and performance monitoring processes have a greater opportunity to leverage the strategic benefit of the FAA.

## **2.5 Ethical Governance, Accountability, and Trust (P2)**

The increasing importance of ethical governance is linked to the fact that FAA systems are used for the development of high-risk strategic decisions (Floridi et al., 2018). Management accounting decisions impact how resources are allocated, how performance is evaluated and what the long-term direction of the organisation will be, thus increasing the ethical risks associated with automation (Lehner et al., 2022).

According to Lehner et al. (2022), the use of AI in making decisions increases uncertainty concerning who is accountable for those decisions, as the recommendations made by an automated system do not clearly delineate the traditional lines of responsibility and therefore undermine governance structures based upon professional judgment and individual accountability, reducing the legitimacy of the decision-making process.

Martin, Shilton and Smith (2020) assert that the appropriate embedding of ethical governance into system design is necessary, rather than simply treating it as an add-on after the fact. There exists a significant risk of losing trust due to bias, lack of transparency, and reduced accountability when organisations focus primarily on efficiency at the expense of ethical considerations (Floridi et al., 2018).

While governance provides a framework of accountability and transparency for the use of AI, there is evidence that over-structured approaches to governance can hinder an organisation's ability to experiment with new technologies such as AI. The complexity and inflexibility of compliance requirements and formal

approvals can create barriers to organisational adaptability and, therefore, delay the potential strategic benefits of FAA adoption (Lehner et al., 2022; Martin, Shilton and Smith, 2020). Therefore, governance must be balanced to achieve both accountability and flexibility to allow organisations to innovate.

As a result, organisations have been adopting more conservative approaches to the implementation of AI-enabled systems, which creates a trade-off in the amount of control an organisation has vs. its ability to experiment with new strategies. Therefore, the relationship between governance and the strategic benefit of using technology is not just about control and flexibility, but rather about how much control you want to give up, allowing your organisation to develop strategically.

Souza et al. (2025) found that increased levels of explainability, audit trails, and human oversight improve both levels of trust and the level of system usage. Conversely, inadequate governance can result in two extremes: (1) complete reliance upon the output of automated systems, or (2) rejection of the system, both of which diminish the quality of the decision-making process (Floridi et al., 2018).

The propositions outlined above provide the basis for Proposition P2, which states that ethical governance moderates the relationship between trust and the adoption of FAA systems.

## **2.6 Professional Development and the Evolving Role of Management Accountants (P3)**

The impact of adopting the FAA will also reshape the way that management accountants define their professionalism in the future. Although automation may reduce some of the work involved in accounting functions (Kokina and Davenport, 2017), it is likely to require management accountants to interpret and advise upon the results of such automated processes (Kokina and Davenport, 2017), rather than simply process the data itself (Quattrone, 2020). The use of digital tools will therefore generate increased levels of informational complexity for management accountants who will need to consider how to contextualise, translate and legitimise the insights generated by such automated tools. This is likely to require an increase in the level of analytical literacy, communications and

strategic awareness required of management accountants to effectively utilise such tools.

Furthermore, while literature highlights the importance of professional transformation, it does so without suggesting that this will be uniform or unopposed; for example, studies indicate that many accounting professionals have resisted automation (Moll and Yigitbasioglu, 2019) due to a fear that the technology could reduce their area of expertise or alter their sense of professional identity. In addition to these barriers to role change, there are additional skill-related constraints (e.g., data analysis, algorithms, systems) which hinder the pace of role transformation; and this is exacerbated when the organisation does not have a formalised training program or capability development process. The literature suggests that the transition to providing advisory services will occur at different rates depending upon the size of the firm/organisation and whether they are constrained by limited resources. Therefore, this body of literature suggests that while professional transformation is dependent on technology being available, it is also dependent upon the organisation investing in capability development and creating an environment that is receptive to the needed changes.

There are empirical studies which provide evidence to support this view. For example, Mahlendorf et al. (2023) demonstrate that using the FAA to improve relationships between senior management and accountants leads to improved strategic planning and evaluation of performance. However, to achieve this benefit, accountants must be able to critically evaluate the results provided by the automated output. Similarly, Rieg (2025) demonstrated that the benefits associated with using dashboards to evaluate performance can only be realised through active engagement with them. Therefore, together these studies support Proposition P3, which highlights the importance of developing professionals as a mediator between the adoption of FAA and creating strategic value.

## **2.7 Dynamic Capabilities, Organisational Agility, and Strategic Responsiveness (P4)**

Dynamic capabilities describe an organisation's ability to detect changes, acquire the necessary assets, and make the necessary changes to its internal



structures to respond to its changing environment (Teece, Pisano and Shuen, 1997).

Several studies have warned that simply investing in digital tools does not automatically create a competitive advantage through dynamic capabilities. Several organisations use advanced analytics but will still not achieve strategic advantages from this investment if they do not develop the necessary supporting organisational routines, align their leadership with these new technology investments and establish collaborative decision-making processes (Csaszar et al., 2024; Warner and Wäger, 2019).

Rieg (2025) also confirmed this finding by demonstrating that real-time dashboard reporting enhanced performance in dynamic environments. However, the relationship between digital technologies and organisational agility is not automatic. While digital investments may provide the potential for generating new dynamic capabilities in organisations (Warner and Wäger, 2019), these can only be achieved through the implementation of appropriate organisational routines, managerial alignment and collaborative decision-making processes (Csaszar et al., 2024).

Inertia (structural), data governance through silos and a reluctance to integrate across functions will inhibit an organisation's ability to convert analytical capability into adaptive action. As such, the contribution of the FAA to dynamic capabilities is dependent on a larger organisational alignment rather than just the technological sophistication.

Collectively, this research offers conditional support for proposition P4 in that the FAA can potentially increase organisational agility when they are embedded into aligned organisational routines and governance systems.

## **2.8 Synthesis, Conceptual Framework, and Analytical Propositions**

The literature indicates that the strategic value of FAA emerges from the interaction between technological capability, governance structures and professional expertise.

While AI-enabled analytics can improve decision quality and organisational agility, the potential benefits from these technologies will depend upon whether there is transparency within the governance environment, as well as the level of professionalism of the staff and the degree to which there is integration of the technology into the organisation. The above synthesis of these findings leads to a proposed conceptual model of how FAA can be used to enhance analytical capacity; the role of ethical governance in moderating the development of trust and the eventual adoption of FAA; the mediating role of professional transformation in creating strategic value through the use of FAA; and the role of dynamic capabilities in enabling organisations to respond strategically to changing environments. It is these relationships that form the foundation for the next set of analytical propositions.

P1: The decision quality is enhanced by the FAA if it is accompanied by human judgment. P2: The relationship between trust and FAA adoption is moderated by ethical governance. P3: The strategic value created by the FAA is mediated by professional transformation. P4: The adoption of FAA may enhance organisational flexibility when embedded within aligned governance and professional capability structures.

To test these propositions, a methodology is required that can synthesise evidence of varying forms and types (i.e., heterogeneous), based on various organisational environments, thus justifying the theory-driven secondary data analysis undertaken in Chapter 3.

## **Chapter 3 – Research Methodology**

### **3.1 Introduction**

This chapter outlines and evaluates the methodological framework used to assess how Financial Automation Analysis (FAA) and Artificial Intelligence (AI) affect strategic decision-making processes in the field of Strategic Management Accounting (SMA). The methodology was specifically designed to meet the research objective and the analytical propositions (P1-P4), outlined in Chapter 1 and Chapter 2, concerning decision quality, ethical governance and trust, professional transformation, and organisational flexibility.

This study adopts a theory-driven secondary data analysis in which documentary sources are treated as empirical evidence and analysed through established theoretical frameworks.

## **3.2 Philosophical Foundations of the Study**

### **3.2.1 Ontological Position: Critical Realism**

The ontological base for this study was based on critical realism. Critical Realism posits that organisational phenomena are independent of the researcher; however, it acknowledges that organisational phenomena can only be imperfectly observed through empirical indicators and social mediation of those observations.

Given that the study of FAA encompasses both technological artefacts (e.g., AI systems, dashboards, predictive models) and organisational outcomes (e.g., decision-making quality, trust, professional roles), a critical realist perspective is most suitable because it recognises that both types of phenomena are "real" but that their impact is dependent on human interpretation and institutional context.

Observable indicators such as forecasting accuracy improvements provide partial evidence of FAA's impact, but strategic outcomes ultimately depend on underlying organisational mechanisms such as governance structures, professional capability and managerial interpretation.

### **3.2.2 Epistemological Position: Pragmatism**

This research will take a pragmatic epistemology (an amalgamation of positivism and interpretivism) because it will rely on quantitative secondary data (observable measures; i.e., changes in forecast accuracy, improvement in decision cycle times, etc.) and qualitative data (narrative accounts and/or descriptions of context) from academic studies, reports, and cases that describe how the FAA impacts or transforms strategic decision making, ethical governance, and professional development.

Positivist approaches focus on measurable relationships between variables, while interpretive approaches emphasise contextual understanding of organisational behaviour.

Therefore, using a pragmatic epistemology will allow us to incorporate multiple methodologies into our research, allowing for a more nuanced understanding of how the FAA is adopted across the industry while providing analytical coherence.

### **3.3 Research Approach**

The research methodology for this study will use a deductive-analytical research design; this involves developing a conceptual framework based upon the findings of the literature review conducted in chapter two, which is informed by three theories: decision theory, socio-technical systems theory and the dynamic capabilities theory. Based upon this conceptual framework, the researcher has developed four analytical propositions (P1-P4); these analytical propositions will be used to guide the empirical analysis of the secondary data.

This type of deductive research does not necessarily entail testing hypotheses using statistical tests but rather acts as an analytical framework that can be used to evaluate secondary data against theoretical expectations and thus determine if, when and to what extent observed organisational outcomes related to FAA adoption have been explained by existing theoretical frameworks.

Due to its ability to integrate secondary data that can be drawn from multiple sources across different organisational settings and methods, secondary data analysis is highly suitable for a deductive-analytical research methodology. An inductive research approach typically requires generating large amounts of primary data, which is a significant limitation of conducting a master's dissertation due to time and resource constraints.

### **3.4 Research Design**

#### **3.4.1 Explanatory Research Design**

The primary method of the current research is an explanatory type of research design. This is because the purpose of the research questions is to explain how and why the FAA has a role in SMA's strategic decision-making as opposed to determining if it does. Explanatory research designs focus on the identification of relationships, mechanisms and contingency contexts that relate to the study objectives.

Descriptive research designs are most appropriate when mapping technology adoptions or technological feature adoptions; however, descriptive designs will not allow the researcher to understand the causal mechanism behind a strategic outcome. Exploratory designs do not have sufficient theoretical grounding to generate meaningful analytic results. Therefore, the explanatory design used in this study is a direct result of a deliberate association between the research problem and methodological approach.

The study uses a secondary data explanatory study design to evaluate how Financial Automation Analysis (FAA) supports the strategic decision-making process of Strategic Management Accounting. This type of design allows for an analytical assessment of empirical as well as professional evidence based on the theoretical framework presented in Chapter 2.

The purpose of this study, therefore, is not to offer a catalogue of all the literature but to discuss epistemologically relevant literature that shows the processes and contingencies of FAA. Therefore, the study does not adopt a formal systematic literature review structure; instead, it applies structured documentary analysis to support theory-driven explanatory research.

### **3.4.2 Cross-Sectional Design**

This research will use a Cross-Sectional Research Design. It will be using Secondary Data primarily from the years 2019 - 2025, supplemented by selected foundational works published before 2019 where theoretically relevant. These years capture recent developments in AI and the FAA's current understanding, and capture differences in organisational experience when adopting technology. Longitudinal studies are better suited for tracking the learning curve and capabilities developing over time. However, most longitudinal studies require the researcher to have long-term access to organisations' private data. Long-term access to an organisation's data is outside the scope of what a master's student can accomplish. Therefore, the Cross-Sectional Study is a practical and academically valid alternative to previous research that has been completed on Digital Transformation in Accounting.

### **3.5 Justification for Secondary Data Analysis**

#### **3.5.1 Nature and Scope of the Method**

Secondary data analysis in this study refers to the systematic analysis of previously collected data obtained for purposes other than the current research. Secondary data analysis is used as a theory-driven explanatory approach to analyse organisational mechanisms and strategic outcomes, and not as a literature mapping activity.

Unlike a systematic literature review, this study adopts a theory-driven secondary data analysis in which selected sources are treated as empirical evidence. Consequently, the chosen literature is treated as empirical evidence to be explained using well-established theoretical models for organisations' mechanisms and strategic outcomes.

Although useful in certain situations, the level of rigour required to explain organisational mechanisms differs from that required to explain individual behaviour. Therefore, the study uses a theory-driven secondary analysis of secondary data that is treated as evidence of an empirical nature and explained through established theoretical frameworks.

#### **3.5.2 Theoretical Justification**

Secondary data analysis allows for engagement with developed theoretical debates and rich evidence from various empirical sources, and to engage with this in a way that would be virtually impossible given the time frame and scope of a Master's project via primary data collection. Through an integration of findings from multiple contexts, the study can analyse both the results observed and the context in which the FAA creates or limits strategic value.

#### **3.5.3 Practical Justification**

Practical factors also give evidence for the utilisation of secondary data. In many cases, access to CFOs, Senior Finance Professionals and Strategic Decision Makers is limited due to both confidentiality issues and time constraints. As a result, collecting primary data through interviewing and surveying may have a major disadvantage in terms of both sample size and respondent bias. By utilising secondary data as an alternative to primary data, the disadvantages listed above

can be minimised because secondary data utilises existing data that has already been systematically collected and verified by reputable organisations. The potential for wider scope, greater feasibility and increased credibility is therefore enhanced by using secondary data while still being in line with Post Graduate Level Expectations.

#### **3.5.4 Methodological Alternatives and Trade-Offs**

Interviews, surveys and mixed-method approaches were considered but rejected due to access constraints and time limitations.

Typically, systematic literature reviews focus on providing a complete picture of research in an area as well as providing replicable information to allow for further studies. The purpose of this study is to provide an interpretive analysis of diverse types of evidence; therefore, it is less concerned with ensuring that the review covers every possible published source. Therefore, it is not well-suited to studying the interplay of the social and technical aspects that make up a socio-technical system.

Secondary data analysis represents a purposeful trade-off from one form of generalisability (statistical generalisability) to another (analytical generalisability). The trade-off that occurs through secondary data analysis is suitable given the explanatory nature of the current study and is consistent with best practices related to using empirical evidence within theoretically based accounting studies.

### **3.6 Data Sources**

To answer the Research Aim and Analytical Propositions (P1-P4), this study has used three types of secondary data, which are: Peer-Reviewed Academic Literature, Professional and Institutional Reports, and Organisational Case Studies. When using these three data types, it allows us to triangulate across theoretical insight, industry-level evidence, and context-specific organisational practice.

#### **3.6.1 Academic Journal Articles: Peer-Reviewed**

Academic peer-reviewed journals serve as the primary source of both theoretical and analytical information for this study. The use of peer-reviewed academic literature is made to:

- define key concepts (i.e., e.g. decision quality, trust, professional transformation) and
- Develop and support the conceptual framework
- Support the development of analytical propositions (P1-P4)
- Provide empirically grounded perspectives on the outcomes associated with the FAA.

Academic literature is highly valued for this purpose due to its methodological rigour and theoretical grounding; therefore, it enables the study to provide a critical evaluation of extant debates concerning accounting and management, as well as those regarding digital transformation. A selection of academic literature was made based on recognition of relevant journals that address issues concerning accounting and management; therefore, this selection ensures scholarly validity. While the use of academic literature is essential to provide a comprehensive understanding of current organisational practices associated with the FAA, such academic literature does not currently capture the scope and scale of many large-scale implementations of AI-enabled financial analysis. As such, many large-scale implementations of AI-enabled financial analysis are documented primarily in professional and institutional reports and not within academic studies. Therefore, academic literature is also complemented by additional data sources to increase the empirical breadth of the research.

### **3.6.2 Professional and Institutional Reports**

Professional and institutional reports present a broad empirical picture of the scale of the adoption, governance, and perceived impacts of FAA amongst institutions and sectors. In general, professional and institutional reports use survey data, benchmarking and practitioner interviews as their primary sources of information and therefore reflect actual practice, which may not have been documented or reported by academics yet.

However, professional reports are approached with caution. The potential exists for a 'positive' reporting bias in the work of consultancies and other institutional bodies when they publish reports. If there is a link between the publication and a consultancy service, then it would be reasonable to assume some degree of positive reporting bias. The limitations associated with positive reporting bias in



professional reports are mitigated by using the findings of professional reports in conjunction with academic research and evidence from organisation cases and therefore treating the findings from professional reports with caution.

### **3.6.3 Organisational Case Studies**

Organisational case studies are useful because they represent real-world examples that illustrate both the practical application of the FAA and how it impacts organisational strategy. The case study process also allows researchers to analyse how the FAA is used in an organisation's process (i.e., how it is integrated into an organisation), how the governance mechanisms are implemented by the organisation, and how individual professionals are affected by automation. Case studies are included because they provide contextual insight into how the FAA is implemented in organisations. While they do not allow statistical generalisation, they enable analytical generalisation by illustrating how theoretical mechanisms operate in practice. Each case is chosen based on whether it addresses the researcher's questions, has credible documentation, and contains enough information to enable a meaningful analysis.

## **3.7 Data Selection and Inclusion Criteria**

### **3.7.1 Inclusion Criteria**

To provide a meaningful secondary source base and to increase the analytic validity and rigour of this research, the literature will include studies that fulfil all four of the following conditions:

- Studies explicitly define their use of AI, automation, or Advanced Analytics within an accounting, finance, or Strategic Decision-Making context.
- Studies contain empirical, analytical, or Theory-Driven insights that relate to at least one of the analytical propositions (P1-P4) developed for this study.
- Studies were primarily published between 2019 and 2025 to ensure they are representative of current trends, with selected foundational theoretical works from earlier periods included where analytically essential.

- Studies originate from respected academic journals, recognized Professional Bodies or Organizations. Therefore, including these studies in the study will allow the studies to contribute to the explanation of the objectives of this study and will not simply describe the studies.

### **3.7.2 Exclusion Criteria**

Sources are excluded when:

- The source is purely about developing a technical algorithm (and does not have an organisational component);
- The source lacks empirical or analytic substance.
- The source represents either speculative or promotional commentary.
- The source does not include any analysis of decision-making, governance, the role of professionals in organisations, or the results of organisational activities. By applying explicit exclusion criteria, we minimise the risk of diluting our analytical findings and ensure that we focus on the most relevant and applicable evidence to support our research objectives.

## **3.8 Source Selection Outcomes and Analytical Mapping**

The method for selecting the documentary evidence was designed according to a structured yet theory-based (as in secondary research) design, as opposed to a formal systematic literature review. The goal was not to find all publications on the FAA but to choose analytically relevant documentation which would be able to help explain the theoretically developed relationships discussed in Chapter 2.

A preliminary review of the literature found that there are 58 possible references. After a process of screening for relevance and assessing quality, 43 references were selected for inclusion in the analysis. The references included within the analysis provided both the theoretical breadth and empirical depth necessary to complete a theory-based secondary data analysis as part of this research.

The final dataset included peer-reviewed academic journal articles, reports from professional associations and institutions, and case studies from organisations. Including all three types of references permitted a form of triangulation by

enabling the comparison of theoretically based findings with practical experience from practitioners and the experiences of individual organisations.

To be able to determine academic breadth, a search was carried out in three databases (Scopus, Web of Science, and Google Scholar) as well as searches for professional bodies' publications. The selection criteria included the relative analytical relevance and theoretical contribution of each source. As such, this represents a form of purposive sampling, which is suitable for developing concepts within interpretative or explanatory secondary data methodologies (Saunders et al., 2019).

In addition, each source was coded thematically along the same four dimensions, which correspond to the four propositions (P1-P4) that had been defined analytically in Chapter 2. Each dimension was used to classify, compare, and map evidence to propositions, and therefore to propositions based upon the most salient conceptual alignment.

Where evidence identified was contradictory, this contradiction was retained to facilitate critical evaluation, rather than being excluded. As such, this process of mapping was both structured and theoretically driven, providing transparency whilst avoiding the rigidities of procedure associated with a systematic literature review, and thus maintaining consistency with the explanatory objectives of this dissertation.

This structured selection process ensured analytical transparency while maintaining flexibility consistent with the explanatory objectives of the study.

### **3.9 Data Extraction Strategy**

The use of a systematic data extraction matrix is implemented to achieve two objectives: each source identified as being relevant for inclusion is examined systematically, and the data that may be relevant to this study are documented using a structured data extraction matrix. The following data will be extracted from each source:

- The publication's context and whether it is an organisational or sectoral context.

- The type of FAA or AI application that was reported to have been used or discussed in the source.
- The degree to which the source reports that the FAA or AI application impacted decision-making, governance, role of the profession, or organisational agility.
- The limitations or caveats identified by the authors of the source;
- The relevance of the source to the analytical propositions (P1-P4).

To enhance methodological transparency and analytical consistency, a structured data extraction matrix was utilised. An illustrative summary of the extraction structure is provided below.

| Source                                | Context                            | FAA / AI Application              | Proposition (P1–P4) | Key Insight                                                                 | Limitation Noted                       |
|---------------------------------------|------------------------------------|-----------------------------------|---------------------|-----------------------------------------------------------------------------|----------------------------------------|
| Fehrenbacher, Ghio and Weisner (2023) | Predictive analytics in accounting | AI-supported forecasting tools    | P1                  | Increased usage of advisory services when outputs are perceived as credible | Model opacity affects interpretability |
| KPMG (2025)                           | Global finance survey              | AI-enabled finance transformation | P2                  | Stronger governance corresponds to greater adoption                         | Potential positive reporting bias      |
| Rieg (2025)                           | Dashboard implementation study     | Real-time performance dashboards  | P3, P4              | Active engagement enhances performance                                      | Cross-sectional design                 |

### 3.10 Data Coding and Organisation

Following on from the extraction process, data are organised and coded within the thematic categories identified by the conceptual framework provided in Chapter 2. The coding process is iterative, allowing for refinement of the themes that emerge through the patterns identified in multiple data sources. The data segments are coded according to one or all the following analytic categories:

- The quality of decision making and professional judgment (P1);
- Ethical governance and professional trust (P2);
- The transformation of professionals and their capabilities (P3);

- The organisational ability to be agile and responsive to strategy (P4). An advantage of the iterative coding process is the opportunity for cross-source comparisons, which allow for the identification of both similar findings (convergent) and different findings (divergent).

### **3.11 Data Analysis Methods**

The analysis of secondary data followed an explanatory research design using a thematic approach as the methodology; the aim of thematic coding was to find patterns and contingency regarding the research aim in the selected sources, as opposed to sequentially summarising the selected sources. Data from selected sources were systematically analysed for conceptual argumentation, empirical evidence, quantitative indicators, and explicit limitations. Thematic coding organised the evidence based on analytical dimensions outlined in the theoretical framework presented in Chapter 2.

These dimensions represented the four focus areas of the study: enhancing decision quality, establishing governance and trust mechanisms, transforming professionals, and increasing organisational agility. Therefore, the coding was informed by the theoretical framework and allowed for changes if emergent themes needed to be clarified or refined. Quantitative indicators provided in professional reports and case studies were viewed as illustrative and non-causal evidence. If a numeric improvement was noted, it was placed within its methodological limits (i.e., based on survey reporting or organisation-specific). This ensured that any numeric claims made about the organisations studied were considered with caution and not generalised beyond the specific evidence supporting them. The analytical process involved a structured progression from coded evidence to theoretically grounded propositions. After thematic coding the selected sources, the extracted segments were organised around the four analytical dimensions that were developed from the literature review. The themes were mapped to propositions P1 – P4 based on the amount of conceptual alignment and the amount of evidentiary consistency. If multiple independent data points yielded similar results, this would be considered additional evidence of the relevant proposition.

In contrast, when conflicting or conditional results were found, those results were preserved and included in the analysis to maintain a balanced perspective. This structured mapping process resulted in the findings contained in Chapter 4 representing systematic analytic aggregation rather than narrative interpretation.

### **3.11.1 Qualitative Thematic Analysis**

The qualitative thematic analysis is the main technique for analysing the findings in this study. Thematic analysis allows researchers to identify common themes, relationships and explanations across several diverse secondary data sets. Unlike descriptive synthesis, thematic analysis in this study is driven by theory as well as conceptually disciplined through being interpreted relative to the analytical propositions, which prevent descriptive drift. In addition to identifying outcomes, the thematic analysis identifies the reasons for the outcomes identified and the conditions within organisations that allow those outcomes to occur.

### **3.11.2 Descriptive Quantitative Comparison**

Where secondary sources report numerical indicators such as forecast accuracy improvements or reductions in reporting cycle times, a descriptive quantitative comparison is used to contextualise qualitative findings.

These quantitative indicators are interpreted illustratively, rather than as evidence of causal relationships. This approach acknowledges the limitations of secondary quantitative data while leveraging its value in supporting explanatory interpretation.

## **3.12 Validity**

The construct validity of the analytical categories developed within this study is supported by the fact that they were grounded within theoretical frameworks established in the literature. Triangulating across three different types of evidence (academic, professional, and case) provides support for internal validity. Rather than attempting to statistically generalise to other populations, external validity is addressed through analytical generalisation.

## **3.13 Reliability**

The reliability of the findings from this study is supported through transparency regarding the data selection, data extraction, and data analysis processes. While

qualitative secondary analysis can never be replicated exactly, the structured nature of the methodology utilised in this study should enhance both the consistency and credibility of the results.

### **3.14 Ethical Considerations**

Although no primary data were collected for this study, ethical issues still apply. All the data that was used in this research is available to the public and properly referenced. It does not contain misleading representations of organisational practices, and it will not make unsubstantiated causal claims about organisations.

### **3.15 Methodological Limitations and Reflexivity**

#### **3.15.1 Methodological Limitations**

This study has several methodological limitations. The use of secondary data limits control over data quality and measurement. Professional reports may contain positive reporting bias, and the cross-sectional design restricts assessment of long-term FAA impacts.

#### **3.15.2 Reflexivity**

Reflexivity recognises that the researcher's own interpretative role in the research process affects how the results of the research are analysed. To limit any potential biases within the analysis, I ground my interpretations in theory and evidence rather than relying solely upon my own judgment. A continued comparison between the data collected during the research, the theories used in the research, and the propositions being tested during the research also serves to maintain some level of analytical discipline.

### **3.16 Alignment Between Research Questions and Analytical Strategy**

The analytical strategy was created to ensure each research question could be answered using a structured review of secondary data as a basis for evaluation. All available evidence relating to forecasting accuracy and analytical capability was mapped to Proposition P1 to evaluate whether automation is related to decision quality. All available evidence relating to governance and accountability was used to support Proposition P2, while all available evidence relating to

professional roles and organisational learning was used to support Propositions P3 and P4. The mapping process ensured that there would be transparency regarding how the evidence informed the analytical conclusions.

### **3.17 Chapter Conclusion**

This chapter has articulated a rigorous, critically informed, and defensible methodological framework for examining the strategic implications of FAA within management accounting. By adopting a theory-driven secondary data analysis approach, the study balances analytical depth with practical feasibility while explicitly acknowledging methodological trade-offs and limitations.

The methodology provides a coherent foundation for the findings presented in Chapter 4 and their discussion in Chapter 5, ensuring alignment with the research aim, analytical propositions, and assessment criteria.

## **Chapter 4 – Findings and Analysis**

### **4.1 Introduction**

These results are based on theory-derived secondary data analysis that examines the strategic use of FAA and AI in Strategic Management Accounting (SMA). The current chapter builds upon the conceptual framework established in Chapter 2; this is accomplished through the interpretation of secondary evidence as empirical evidence for the evaluation of the analytical propositions using qualitative thematic analysis and illustrative quantitative evidence.

Therefore, the purpose of this chapter goes beyond reporting on the outcomes associated with FAA adoption; it examines how and the organisational circumstances under which FAA supports quality in strategic decision-making, ethical governance, trust, transformation of professionals, and organisational agility. Findings will be organised according to the four analytical propositions (P1-P4).



## **4.2 Financial Automation Analysis, and Quality of Decision Making in Strategic Management Accounting (P1)**

### **4.2.1 Improved Analytical Capabilities and Enhanced Strategic Insight**

The evidence collected through analysis indicates that financial automation analysis improves analytical capabilities in strategic management accounting by enabling the ability to process data in real time and perform predictive modelling.

Evidence from professional reports, for example, Deloitte (2025) and the ACCA (2024), indicates that there is an evolving organisational trend from traditional static annual budgeting toward ongoing or continuous forecasting and rolling scenarios.

### **4.2.2 Quantitative Indicators of Decision Quality Improvement**

Reported quantitative indicators are interpreted illustratively rather than causally, reflecting heterogeneous organisational contexts, measurement approaches, and implementation maturity levels across studies.

Multiple studies report quantitative measures of improved analytical capabilities because of adopting financial automation analysis. According to reports from ACCA (2024) and Deloitte (2025), forecasting systems powered by artificial intelligence enhance accuracy by around 10 to 25%, which demonstrates a high level of consistency with other studies examining the use of predictive analytics within accounting (Fehrenbacher, Ghio and Weisner, 2023). Additionally, studies have demonstrated that continuous forecasting models have resulted in a reduction of up to 20% in budgetary variance from traditional methods of forecasting. It is important to approach these figures with caution, viewing them as suggestive patterns rather than definitive proof of causation.

Organisational cases show that improved forecasting accuracy supports better inventory planning and pricing decisions.

### **4.2.3 Limitations of Automation and the Role of Human Judgement**

The improvements identified in the studies above notwithstanding, all the studies found that the use of FAA (as with any other technological innovation) will never fully eliminate the necessity for a professional's judgment in making business decisions. The literature concerning FAA also emphasizes caution regarding

what is referred to as “technology dominance” (Arnold and Sutton, 2019) which occurs when users rely too heavily on the output of an automated system such that they reduce or eliminate their own analysis and understanding of context; however, the literature also suggests that those organizations that have achieved the largest improvements in decision quality view FAA as a decision support tool, not as a decision maker. The case evidence presented provides further support for this interpretation. As indicated in the literature, in practice, management accountants remain responsible for analysing the strategic implications of the insights provided by an FAA, as well as considering the relevant ethical issues and context before reaching a final decision (Verhoef et al., 2021).

The evidence supports Proposition P1, indicating that FAA improves decision quality only when combined with human judgment. However, several studies report limited decision-quality improvements where data governance was weak or managerial engagement with automated outputs remained superficial.

These findings align with the conditional interpretation advanced in Chapter 2, where the strategic value of FAA was theorised as dependent on governance maturity and professional capability.

## **Ethical Governance, Trust, and Adoption of FAA (P2)**

### **4.3.1 Governance Challenges in AI-Enabled Accounting**

The findings suggest that ethical governance will be a key factor in whether there is an increase in trust and the use of FAA within SMA. As AI-generated insights become more prevalent in high-stakes decision-making, the need for increased accountability, transparency, and responsibility becomes more apparent.

The analysed evidence indicates that accountability ambiguity represents a key governance challenge when organisations implement FAA. In many cases, the accountability for outcomes can become diffused among system developers, data providers, and system users, creating challenges for existing governance frameworks that rely on individual professional accountability. Additionally, the common risks associated with AI include algorithmic bias, data integrity issues, and the inability to provide explanations for the systems' recommendations have been identified as some of the major barriers to the successful implementation of the technology.

#### **4.3.2 Quantitative Evidence Linking Governance and Adoption**

Quantitative evidence supports the role of governance mechanisms in facilitating the effective application of professional judgment when using Artificial Intelligence (AI) generated insights. Professional surveys revealed that many senior finance professionals were unwilling to utilise AI-generated insights without formally defined governance structures to govern the utilisation of these systems.

In contrast, organisations implementing a formal AI governance policy report a significant increase in the number of users of FAA tools compared to organisations lacking a formal AI governance policy. Organisations that implement transparent governance mechanisms (for example, using Explainable AI and Audit Trails) tend to have greater usage rates for AI-enabled Decision Support Systems than organisations that do not implement such governance mechanisms (Souza, de Araujo Wanderley and Braga de Aguiar, 2025; KPMG, 2024; Ranjith et al., 2021).

Collectively, these quantitative results illustrate that, for organisational stakeholders to be able to trust in the use of FAA systems, there must be appropriate governance structures that ensure accountability and interpretability.

#### **4.3.3 Governance as an Enabler of Strategic Use**

The evidence supports Proposition P2, indicating that ethical governance moderates the relationship between trust and FAA adoption. Processes of governance can add to the procedural complexity of testing with AI-based systems, which will be slower when there is no additional support for the organisational culture and capability in terms of the governance framework (Lehner et al., 2022; Souza, de Araujo Wanderley and Braga de Aguiar, 2025). Therefore, governance frameworks alone are often insufficient to change organisational behaviour or adoption patterns.

## **4.4 Professional Transformation and Strategic Value Creation (P3)**

### **4.4.1 Transformation of the Management Accounting Function**

The research found that the management accounting role has consistently changed since the adoption of FAA technology. In many cases, automation of routine analytical work has taken over the accountants' time for advisory work, strategic work, and making sense of data generated by automated tools. This shift reflects the emergence of hybrid management accounting roles combining analytical, strategic and communication capabilities.

The academic view of the shift from traditional accounting practices to those which incorporate artificial intelligence is that this represents an increase in information complexity and, therefore, requires an accountant to interpret and place into context the AI-generated insight (Raisch and Krakowski, 2021; Quattrone, 2020).

### **4.4.2 Quantitative Evidence of Role Transformation**

Quantitative measures of performance are a strong indicator of changes, such as work that occurs when the profession is transformed through the introduction of the FAA. Reports from Professional Bodies indicate a decline in Time Spent on Routine Reporting and an associated Increase in Time Devoted to Strategic Analysis and Business Partnering Activities (Deloitte and Institute of Management Accountants, 2024; ACCA, 2024). It has also been reported by organisational case studies that productivity improvements exist in finance departments after the use of AI-based controllership practices.

### **4.4.3 Capability Development as a Mediating Mechanism**

Importantly, this research illustrates that professionals must develop capabilities for professional transformation through an investment of time in capability development. Organisations which fail to invest in analytical training and capability development may find it difficult to achieve the strategic advantages of adopting FAA (Rieg, 2025; PwC, 2025).

The constraint is whether users will utilise the analytical capabilities of the FAA at a level of sophistication greater than superficial use, or if they can interpret the

results of the analysis (and therefore not result in strategic benefit) or misinterpret the automated insights and thus limit the amount of value created.

Therefore, these findings confirm that professional transformation mediates the relationship between the adoption of FAA and the creation of strategic value as stated in Proposition P3.

Role transformation was uneven across organisations, particularly where investment in analytical training and cross-functional collaboration was limited.

## **4.5 FAA, Dynamic Capabilities, and Organisational Agility (P4)**

### **4.5.1 Strengthening Dynamic Capabilities through Advanced Analytics**

Findings show that the use of FAA enables organisational agility through the development of a dynamic capability to sense, seize, and reconfigure strategically relevant opportunities. AI-enabled analytical tools facilitate faster detection of emerging trends within organisational and market data for an organisation to capitalise on compared to traditional reporting methods (Csaszar et al., 2024; Warner and Wäger, 2019).

### **4.5.2 Quantitative Evidence of Agility Improvements**

Quantitative data demonstrates a significant reduction in reporting cycle time after the FAA was implemented. The OECD, BCG, and INSEAD (2025) highlight instances where the duration of reporting cycles was shortened from weeks to days due to the integration of AI technologies; there is also documentation that faster strategic decision-making has taken place due to the contributions of the FAA for improved responsiveness.

The organisational case studies demonstrate how near real-time analytics allowed for rapid adjustments to the company's supply chain strategy in response to changes in the demand conditions, therefore illustrating that FAA's value proposition can be for strategic agility as well as operational efficiency.

### **4.5.3 Continuous Learning and Strategic Adaptation**

Automated dashboards and real-time analytical tools allow management accounting professionals to continually assess an organisation's performance against its strategic objectives (Rieg, 2025), thereby allowing their organisation to continue to develop its ability to adapt to changes in its environment over time.

These findings provide consistent evidence supporting Proposition P4, as they illustrate that the FAA facilitates an organisation's ability to be more agile by increasing its dynamic capabilities. Moreover, faster decision-making did not necessarily equate to better strategic clarity, suggesting that agility necessitates coordinated organisational alignment and not merely acceleration.

#### 4.6 Interdependence of Analytical Propositions

An important finding to emerge from this analysis is the interrelationship between P1-P4. Improvements in decision quality (P1) are contingent upon trust in automated outputs, which is shaped by the presence of robust ethical governance structures (P2). Through effective governance and trust, professionals can engage with their work at a deeper level and transform themselves (P3). This professional transformation enables organisations to achieve greater levels of agility (P4). Therefore, the systemic relationship demonstrated above illustrates that the FAA adoption process is best viewed as an organisational transformation process and not simply a discrete technological implementation. Therefore, the strategic value is derived from how technological abilities, governance frameworks, professional knowledge, and the organisational environment interact with one another, which are all consistent with what is referred to as a socio-technical view of digital transformation (Jarrahi, 2019; Quattrone, 2020; Verhoef et al., 2021).

#### 4.7 Quantitative Synthesis of Key Findings

The table provides an overview of some of the typical trends found in the various studies, but does not represent a statistical compilation of standardised effect sizes.

| Dimension                   | Indicative Outcome                   | Source                               |
|-----------------------------|--------------------------------------|--------------------------------------|
| Decision quality            | 10–25% forecast accuracy improvement | ACCA (2024); Deloitte (2025)         |
| Governance & trust          | 30–40% higher AI utilisation         | KPMG (2024)                          |
| Professional transformation | ~50% reduction in manual reporting   | ACCA (2024); Deloitte and IMA (2024) |

| Dimension              | Indicative Outcome            | Source                      |
|------------------------|-------------------------------|-----------------------------|
| Organisational agility | 20–35% faster decision cycles | OECD, BCG and INSEAD (2025) |

Reported ranges should therefore be interpreted as indicative patterns observed across multiple studies rather than as precise effect sizes attributable solely to FAA adoption.

Organisational context variation, data infrastructure quality, and governance maturity significantly affect observed outcomes.

Collectively, the findings discussed above support the analytical propositions established in Chapter 2.

- Proposition P1 is supported by evidence demonstrating that forecast accuracy improvements and enhanced analytical depth are contingent upon active human oversight and professional judgment.
- Proposition P2 is further reinforced by findings indicating that governance frameworks moderate the relationship between trust and FAA adoption.
- Proposition P3 is evidenced through documented professional role transformation and capability development as mediating mechanisms of strategic value creation.
- Proposition P4 is supported by both qualitative and quantitative evidence showing enhanced organisational agility, decreased reporting time lag, and quicker strategic response cycle times

## 4.8 Chapter Summary

This chapter explored the strategic implications of the FAA for SMA by utilising a theory-based approach to analyse secondary data. The results demonstrated empirical evidence supporting each of the analytical propositions and demonstrated how the strategic value of the FAA arises from incorporating the FAA into governance systems, professional capabilities, and organisational processes.

Building upon these findings, the next chapter will summarise the conclusions from this study, detail the implications of the study's findings for practitioners and researchers, and suggest directions for future studies.

## **Chapter 5 – Conclusions and Recommendations**

### **5.1 Introduction**

This chapter summarises the findings of the theory-driven secondary data analysis of FAA and AI in Strategic Management Accounting.

The chapter will revisit the research aim, objectives and analytical propositions (P1-P4) based on the findings from Chapter 4, and identify the theoretical, practical and professional implications arising from the study.

### **5.2 Addressing the Research Aim and Questions**

The overall objective of this dissertation is to explore the effect that Financial Automation Analysis and Artificial Intelligence have on strategic decision-making in the area of Management Accounting using secondary data; this overall objective has been achieved by combining academic literature, industry reports, and organisational case studies with secondary data as a basis for analysis through the use of three theoretical lenses (Decision Theory, Socio-Technical Systems Theory, Dynamic Capabilities Framework).

**Research Question 1: How do the results of Financial Automation Analysis (FAA) improve the quality and effectiveness of strategic decision-making within Strategic Management Accounting?**

The results show that FAA provides better-quality, more relevant and timely information to support the strategic decision-making process in terms of providing ongoing forecasts, allowing for analysis of scenarios and combining both financial and non-financial data. However, the extent to which the quality of decision-making will be enhanced depends on how it is utilised within an organisation.

Thus, there are conditions under which the FAA will provide better decision-making. When FAA is used in conjunction with managerial judgment, the quality of decisions made will be greater. Additionally, when managerial accounting staff utilise their professional judgment and provide context regarding interpretations of the outputs from FAA, then decision-making will also be improved. The latter supports proposition P1 and supports the idea that automated tools, such as FAA, will enhance managerial decision-making but will never replace it.



This interpretation aligns with the bounded rationality perspective outlined in Chapter 2.

**Research Question 2: What ethical and governance issues arise when there is increased reliance on AI-driven financial analysis?**

The results show that ethical governance is important for establishing trust in FAA systems. Secondary sources show that issues with accountability, transparency, and explainability were continually present. Where the governance structure was weak or non-existent, organisations experienced either resistance to adopting FAA systems or an over-reliance on the automated output from these systems.

Both practices, as they relate to FAA adoption, reduce the ability of an organisation to make strategic decisions effectively. The findings establish that ethical governance moderates the relationship between trust and the successful adoption of financial automation analysis and thus supports Proposition P2. These findings support the governance–trust relationship described within the socio-technical framework.

**Research Question 3: In what ways are the roles and capabilities of management accountants developing because of FAA adoption?**

The findings demonstrate an association between FAA adoption and the transformation of the management accounting role. The automation of routine reporting tasks frees up time for the management accountant to be involved to a greater degree with strategic analysis, business partnering and advisory functions.

However, this transformation depends on investment in the analytical, interpretative, and communication competencies required to leverage the potential strategic benefits from the use of FAA. Without such investment, the strategic value of the FAA is limited.

Therefore, the research findings provide considerable convergent evidence to support Proposition P3, which indicates that professional transformation is a critical mediating mechanism in the relationship between FAA adoption and strategic value generation.

This interpretation reinforces the mediating role of professional transformation discussed in Chapter 2.

**Research Question 4: In what way can the FAA contribute to an organisation's ability to be responsive and agile, and its strategic performance?**

The evidence shows that the use of FAA (by organisations) will contribute to organisational agility through enhancing an organisation's capacity to identify (sensing), exploit (seizing), and adjust to new (reconfigure) strategic opportunities, enabling rapid real-time responses to changing circumstances through a combination of real-time performance monitoring and fast reforecasting.

Furthermore, the benefits of using FAA for improving strategic performance are indirect and dependent upon the effectiveness of governance structures and the level of professionalism amongst those who use it. Therefore, this provides support for Proposition P4 in demonstrating that the FAA can enhance organisational agility when used as part of an overall set of organisational capabilities.

These results are consistent with the dynamic capabilities perspective applied in Section 2.7.

**Research Question 5: Which of the various analytical frameworks is most effective at explaining the inter-relationship between FAA, governance, decision quality and organisational capabilities?**

The findings support the utility of an integrated analytical framework developed within Chapter 2. Decision theory supports how the FAA addresses bounded rationality; Socio-Technical Systems Theory provides context for the changes that occur in professional and organisational practices; The Dynamic Capabilities Framework shows how the FAA is contributing to strategic responsiveness. These three frameworks explain why the FAA produces disparate results among organisations.

### **5.3 Theoretical Contributions**

This study integrates decision theory, socio-technical systems theory and dynamic capabilities to explain how Financial Automation Analysis influences

strategic decision-making. The study builds on previous research that has studied decision-making processes under the influence of technological changes (as part of AI) through the integration of three frameworks: (1) Decision Theory; (2) Socio-Technical Systems Theory; and (3) Dynamic Capabilities Framework into one explanatory structure to better understand how technological capability, governance and professional practice are interdependent factors that ultimately affect organizational strategy.

The results expand socio-technical systems theory by showing how professional roles and organisational practices moderate the strategic value created from automation technologies. This study also contributes to the dynamic capability literature in that it demonstrates how the use of financial analytics through AI may assist organisations to sense, seize and reconfigure, which are the three main processes that lead to an organisation's strategic agility.

By combining these three different perspectives, this study develops a better understanding of how the FAA can influence governance structures, professional roles and decision quality.

## **5.4 Practical Implications for Organisations**

### **Short-Term (0–12 Months) – CFO / Finance Director**

- Establish formal AI governance frameworks, including explainability standards, audit trails, and defined accountability structures.
- Position the FAA explicitly as a decision support mechanism, and not a decision substitution mechanism.
- Carry out a governance risk assessment before implementing it at scale.

### **Medium-Term (1–3 Years) – Finance Leadership & HR**

- Invest in structured capability development programmes focusing on data literacy, analytical interpretation, and strategic communication.
- Embed FAA into core management accounting processes, including budgeting, rolling forecasting, and performance management.
- Develop cross-functional collaboration mechanisms to integrate financial analytics with operational decision-making.

## **5.5 Professional Implications for Management Accountants**

The results demonstrate that there is a need for updated professional skills by the management accounting profession due to automation of the majority of the FAA's analytical work. Management accountants will have to transition from being analytical technicians and will be required to develop into strategic consultants that link automated analysis with organisational decision-making. Therefore, it would be advantageous for professional organisations and those involved in providing education/training for management accountants to focus on developing their members' knowledge/skills in data literacy, analytical interpretation, ethical decision-making, and strategic communication. If the professional organisations do not provide adequate support for updating the skills of management accountants, this could result in the loss of the strategic value of the management accounting role in an increasingly automated business environment.

## **5.6 Methodological Reflections and Limitations**

This study uses a theory-driven secondary data analysis methodology that has some advantages (for example, it provides access to large amounts of empirical data from multiple organisations) but also some disadvantages. Firstly, the use of secondary data limits control over how data is collected or measured. Secondly, there could be positive reporting biases within the professional reports used for data collection, and therefore, they need to be interpreted cautiously. Thirdly, the secondary data is cross-sectional rather than longitudinal, so it does not provide information about the longer-term implications of adopting an FAA. The limitations have been addressed using triangulation, critical evaluation, and explicitly acknowledging these issues. The methodological trade-offs that were made were appropriate given the research objectives and the constraints placed by conducting a master's dissertation.

As a result, the conclusions of this study are analytically generalisable rather than statistically generalisable, contributing to theory development and explanatory insight rather than predictive quantification.

## 5.7 Recommendations for Future Research

Future studies can expand on this research as follows. A longitudinal design that tracks how the FAA develops its capabilities will help to understand the impact of experience (learning) and how it affects the ability to develop capabilities. In addition to primary data obtained through a survey of finance professionals about their governance practices and professional judgments, a greater understanding of these topics is likely to be achieved through interviewing finance professionals. Finally, comparative studies across different industries or countries may show how the environment in which an institution operates affects FAA outcomes. Furthermore, additional research could be conducted on the interactions between the FAA and new regulatory frameworks that are developing to govern the use of artificial intelligence in making financial decisions.

## 5.8 Final Conclusion

This study shows that the FAA can improve strategic decision-making in management accounting, but its value depends on governance, professional capability and organisational context.

This study used theory-driven secondary data analysis to examine the FAA as a socio-technical transformation. The results show that for organisations to achieve the strategic benefits of automation, they will need to invest in both the technical systems and their governance and professional development capabilities. Overall, FAA supports more informed strategic decision-making when supported by appropriate governance and professional capability.

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## Appendices

### **Appendix A: Ethical Considerations Statement**

This study was completed in line with Brunel University London's ethical framework. Since the research was reliant entirely on secondary resources (i.e., academic peer-reviewed journals, organizational case studies, institutional reports), no primary data collection occurred and there were no human subjects involved at any point during the research process.

As all secondary data sources used within this study are publicly available, appropriate citation and reference using the Harvard format has been applied to each resource. The nature of the study is such that it does not collect, store, or process individual data; therefore, it poses no issues relative to the General Data Protection Regulations (GDPR) or the Data Protection Act (2018).

Findings, interpretations, and conclusions developed through this study are based entirely upon an examination of secondary data and do not contain any false information about the practices of organisations studied, nor are there any unsubstantiated causal claims being proposed. All third-party materials were treated as intellectual property, and proper attribution was provided to each source.

The researcher states that this study was conducted ethically and in compliance with Brunel University London's Research Ethics Framework. Because the study

utilized secondary resources exclusively, a formal ethics review by the University Research Ethics Committee was not required.

## Appendix B: Data Source Extraction Matrix

The following matrix provides an illustrative summary of the data extraction framework applied to key sources utilised in this study (as referenced in Section 3.9). Each source was coded against five analytical dimensions aligned with the research propositions P1–P4.

| Source                         | FAA/AI Application                     | Decision Impact                          | Governance/ Ethics                    | Role/ Agility                    | Propositions (P1–P4) |
|--------------------------------|----------------------------------------|------------------------------------------|---------------------------------------|----------------------------------|----------------------|
| Deloitte (2025)                | Predictive analytics; NLP reporting    | High improved forecast accuracy –        | Moderate risk frameworks referenced – | Significant role shift noted     | P1, P3, P4           |
| KPMG (2024)                    | AI-enabled financial dashboards        | High – real-time strategic insight       | High – AI governance frameworks       | Finance function transformation  | P1, P2, P3           |
| Giantsidi & Tarantola (2025)   | Deep learning for forecasting          | High accuracy improvements quantified –  | Low – limited governance discussion   | Not addressed                    | P1                   |
| Lehner et al. (2022)           | AI decision-support systems            | Conditional – human oversight required   | High accountability structures –      | Accountant roles redefined       | P1, P2, P3           |
| Rieg (2025)                    | Digital tools in management accounting | Moderate – context dependent             | Moderate                              | Capability development discussed | P3, P4               |
| Sainsbury's / Microsoft (2024) | Azure AI supply chain tools            | High – agility improvements reported     | Moderate – internal governance        | Cross-functional collaboration   | P1, P4               |
| Raisch & Krakowski (2021)      | AI–human complementarity model         | Conditional – augmentation vs automation | Moderate                              | Strategic role of human judgment | P1, P3               |
| Martin et al. (2020)           | Ethical frameworks in business         | Indirect                                 | High – ethical embedding in systems   | Professional responsibility      | P2                   |
| PwC (2025)                     | AI jobs barometer; finance automation  | High productivity gains reported –       | Moderate                              | Significant workforce impact     | P3, P4               |
| OECD et al. (2025)             | AI adoption in firms                   | High – cross-sector evidence             | High regulatory and governance –      | Dynamic capabilities discussed   | P2, P4               |

### **Appendix C: Artificial Intelligence Use Declaration**

In compliance with Brunel University London's Academic Integrity Policy, I declare that Artificial Intelligence (AI) tools were used in the preparation of this dissertation as follows:

Tool(s) used: Claude (Developed by Anthropic)

How it was used:

- Editing of the format/structure of the dissertation document (including font, spacing, margins & headings)
- Reference checking/correction to ensure that all citations/references in the document are compliant with the requirements of the Harvard referencing system
- Checking for grammatical errors and statements that may be incorrect in the written material

Explanations of how the AI tool was not used:

The researcher generated and wrote ALL intellectual content of this dissertation (including the research aim, analytical propositions, theoretical framework, literature review, methodology, findings, analysis and conclusion)

by themselves. The researcher did not use AI tools to create, rewrite or structure the substantive academic material presented in this submission.